

(a) Observation Budget Parity

Strict $N=2$ Satellite Budget:

- Exactly 2 SLC acquisitions per track
- 12-day repeat ($\Delta t = 12$ d, C-band 5.4 GHz)
- Dual-pol: VV + VH complex channels

4 Satellite Tracks (Cross-Track Range)

T15	T37	T88	T139
Asc.	Desc.	Asc.	Desc.
32.8°	37.1°	41.4°	45.0°

Angle disparity: $\Delta\theta_{\max} = 12.2^\circ$ on common support Ω

(b) Five-Level Representation Ladder

I_8	8-D	2-date intensity baseline (VV, VH, ratios)
I_{28}	28-D	I_8 + multi-scale textures & gradients
$I_8 + g_{12}$	9-D	I_8 + raw 12-day scalar coherence γ_{12}
F_{12}	12-D	$I_8 + g_{12}$ + 3 derived transforms (texture, grad, log)
$I_{28} + g_{12}$	29-D	Full fusion: 28-D intensity + scalar coherence

Physical scalar $\gamma_{12} \in [0, 1]$ introduces phase stability

(c) Spatial Validation Guard

Strict Autocorrelation Control:

- 40 Spatial Blocks (5×5 km grid)
- 28 Training Blocks (20,000 pixels)
- 12 Held-out Evaluation Blocks (9,600 px)

Spatial Isolation Guarantees

- 400 m Buffer Guard**
k-d tree distance floor between train & val
- 3-Fold Spatial CV**
North, Central, South source partitions
- 5.0 km Holdout**
Directional North buffer robustness check

No feature-window overlap; residual correlation bounded

(d) Cross-Track Transfer Evaluation & Statistical Inference Framework

12 Cross-Track Pairs ($s \neq t$)

- Train on source track $s \in \{15, 37, 88, 139\}$
- Evaluate on target track $t \neq s$
- $4 \times 3 = 12$ directional transfer tasks
- Evaluates cross-incidence & cross-heading
- Strict held-out spatial test blocks

Primary Estimands & Gap Formulation

$$\text{BA}_{\text{pooled}} = \frac{1}{12} \sum_{s \neq t}^5 \sum_{k=1}^5 \text{Recall}_{s \rightarrow t}(k)$$

Primary Metric: Unweighted macro-average across 5 classes

Transfer Gap: $G = \text{Acc}_{\text{in}} - \text{Acc}_{\text{cross}}$

Gap Compression: $\Delta G = G(I_{28}) - G(I_{28} + g_{12}) = \Delta_{\text{cross}} - \Delta_{\text{in}}$

Joint Block Bootstrap Inference

- $B = 20,000$ joint spatial block bootstrap resamples
- Shared block resample indices across all arms
- Complete-class replicates: 19,306 valid draws
- Nonparametric percentile 95% confidence intervals
- Paired 12-block exact sign test ($p = 0.0005$)

Saved Models, Predictions & Verification Logs